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Nebius x NVIDIA Global AI Hackathon · Physical AI track · simulated depot

Fleet Charge Controller

A scheduler and edge-control system that charges a 50-truck electric depot through 11 shared bays. Each truck gets exactly the energy its trip needs, at the gentlest and cheapest rate that still gets it to the gate before dispatch.

39%
lower daily cost than first come, first served

27%
lower peak grid demand

50 trucks · 11 bays
three model pools, plug-in and unplug delays

0.00 / day
missed departures, 8 days averaged

This brief documents the whole project: the problem, the models, the math, the scheduler, the simulator, the language-model layer, the measured results, and what is still missing. Every number in the charts and tables comes from the project's own simulator, run from the code in the repository. Figures marked *synthetic* use generated prices, solar, arrivals and trips.

HOW TO READ THE RESULTS

Costs are for one simulated day of a 50-truck depot under a made-up but plausible tariff. They show how the policies rank against each other. They are not a forecast of any real site's bill.

1 · REQUIREMENTS

The problem

A depot operator has trucks that return at different times, must leave at different times, and need different amounts of energy for the next trip. Electricity costs change through the day. Solar panels give free energy at midday. Chargers are scarce: there are fewer bays than trucks, and a Ford E-Transit cannot use a bay built for a Mercedes eActros. Fast charging is cheap in time and bad for the battery.

The controller has to answer four questions continuously:

1. **How much** energy does each truck need? It depends on distance, cargo weight, terrain, temperature, the battery's age, and a safety buffer.
2. **Which truck** takes which bay, and when? Trucks wait in a lot for a bay of their own model.
3. **How fast** should each truck charge? Slowly when there is slack, so it costs less and wears the battery less. Quickly when the deadline is close.
4. **Is everyone ready** at dispatch time, with the connector off, the truck at the gate and a margin to spare?

REQUIREMENT	WHERE IT LIVES
Dynamic pricing	Price per slot enters the planner's objective (§8)
On-site solar	Solar forecast offsets grid import in the site balance; realised solar differs from the forecast and the edge loop corrects (§8, §10)
Different arrival times	Each truck's release slot; queues form when bays are full (§9)
Different dispatch times	Each truck's deadline, with a 15-minute dispatch buffer (§9)
Charge only what the journey needs, plus a buffer	Physics energy model with margin and reserve (§4)
Cargo weight and distance	Mass and distance terms in the energy model (§4)
Battery health	Charge gently, avoid sitting at high SOC; SOH changes the usable capacity and the power limit (§5, §7)
Fewer bays than trucks, separate pools per model	Bay scheduler (§9)
Delays moving into a bay and out to dispatch	Plug-in and unplug overhead in the schedule and in the simulator (§9, §10)
Trucks report SOC and SOH	Noisy telemetry the controller plans from (§5)

2 · INPUTS

Fleet and site

The fleet uses three real truck models. Battery, range and charging figures come from public manufacturer and press sources (checked October 2026). Mass, drag and drivetrain values are engineering estimates for the energy model, and the model is then calibrated to the published range (§4).

MODEL	COUNT	BATTERY (USABLE)	RATED RANGE	CHARGING	BAYS	DEPOT CHARGER
Ford E-Transit (2025)	28	89 kWh	143–159 mi	19.2 kW AC (DC up to 176 kW)	6	19.2 kW
Volvo VNR Electric	14	565 installed, ~452 usable	up to 275 mi	250 kW CCS1	3	150 kW
Mercedes-Benz eActros 600	8	621 installed, ~590 usable	500 km	up to 400 kW CCS2	2	180 kW

Sources: Autoblog on the 2025 E-Transit, InsideEVs on the VNR Electric, InsideEVs on the eActros 600.

The site

- Grid import limit 1,500 kW; solar peak 900 kW; charger efficiency 93 % (grid kWh to battery kWh).

- Price: a time-of-use base (0.09 $/kWh overnight, 0.14 daytime, 0.32 from 16:00 to 21:00) with hourly noise and one real-time spike of 1.8x in the evening peak.
- Demand charge: 0.55 $ per kW of the day's peak grid import, a prorated monthly tariff.
- The day runs from 12:00 to 12:00 in 96 slots of 15 minutes. Arrivals are 12:30–22:30 (peak about 17:30); departures are 04:30–12:00 next day. About 18 % of vans make a same-day short turn.

The fleet's spread

Every truck has its own random arrival SOC (15–45 %), its own true SOH (82–100 %), cargo between 15 % and 100 % of the model's maximum payload, terrain between 1 and 8 m of climb per km, and a priority (15 % of trucks count double when they miss a target). If a generated trip would need more than 95 % of the battery, the route is shortened by 4 % steps until it fits, as a dispatcher would.

3 · STRUCTURE

Architecture

The design rule: **language models read and explain, deterministic optimisation decides**. Deadlines must hold, so no model sets a power level or picks a bay. The system works on two time scales: a slow planning loop over 15-minute slots and a fast edge loop every minute.

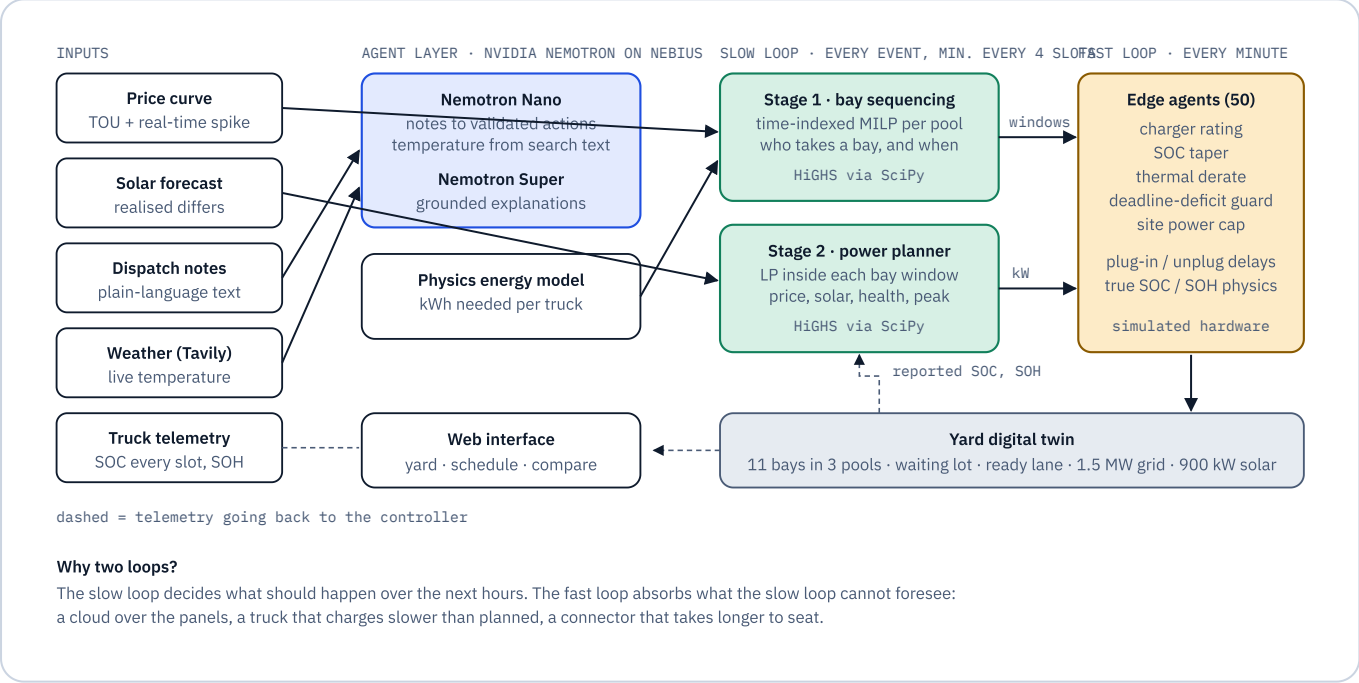


Figure 1. The controller. Optimisation (green) owns bay assignment and power; Nemotron (blue) owns language; edge agents (amber) own the last-minute safety limits.

Why this fits the Physical AI track

The track asks for agents that sense and act in the physical world, with serverless jobs used to evaluate policies at scale. This project has a sense-decide-act loop around simulated hardware: telemetry in (SOC, SOH, temperature, plug state), setpoints out (kW per charger, bay assignments), local safety limits at the edge, and a large benchmark of policies run as a batch job. There is no physical hardware yet, so the video will show the key modules in action, as the rules allow.

Trip energy model

A flat "kWh per km" ignores the things a dispatcher knows matter: a loaded tractor-trailer burns a lot more than an empty one, a hilly route costs more than a flat one, and cold weather costs a lot on every truck. The model builds the trip's battery energy from physics and then scales it once per truck model so that, at reference conditions, it reproduces the consumption implied by the published range.

Forces and losses

$m = m_{\text{curb}} + m_{\text{payload}}$	mass
$E_{\text{roll}} = C_{\text{rr}} \cdot m \cdot g \cdot d$	rolling resistance
$E_{\text{aero}} = \tfrac{1}{2} \cdot \rho \cdot C_d A \cdot (1.05\,\bar{v})^2 \cdot d$	aerodynamic drag
$E_{\text{prop}} = (E_{\text{roll}} + E_{\text{aero}}) / \eta$	battery to wheel
$\kappa = 1/\eta - \eta \cdot r$	net cost of one joule of motion energy, with regeneration fraction r
$E_{\text{stop}} = n_s \cdot \tfrac{1}{2}\, m\, v_s^2 \cdot \kappa$, $n_s = (\text{stops per km}) \cdot d$	stop-and-go
$E_{\text{climb}} = m \cdot g \cdot h \cdot \kappa$, $h = (\text{ascent per km}) \cdot d$	climbing, net of descent
$P_{\text{hvac}} = k_h \cdot \max(0, 18 - T) + 0.6\, k_h \cdot \max(0, T - 24)$	heating and cooling, T in °C
$E_{\text{aux}} = (P_{\text{aux}} + P_{\text{hvac}}) \cdot d / \bar{v}$	auxiliaries over the trip time
$f_{\text{cold}} = 1 + 0.004 \cdot \max(0, 15 - T)$	cold battery and tyres
$E_{\text{trip}} = c \cdot f_{\text{cold}} \cdot (E_{\text{prop}} + E_{\text{stop}} + E_{\text{climb}}) + E_{\text{aux}}$	battery energy for the round trip

d is the round-trip distance, $\bar{v}$ the average speed, g = 9.81 m/s², ρ = 1.2 kg/m³. Energies in joules convert to kWh by dividing by 3.6 million.

Calibration

The constant c is solved per model so that at reference conditions (100 km, half the maximum payload, 3 m/km ascent, 15 °C) the model returns the consumption derived from the published range, $\hat{c}$ = usable kWh ÷ rated range:

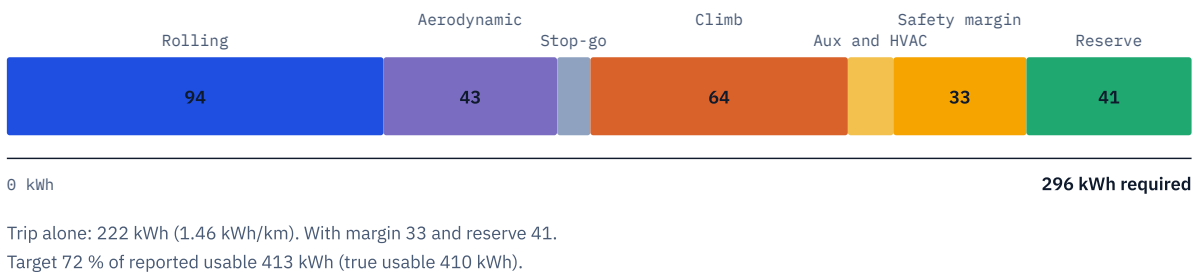
$$c = (\hat{c} \cdot d_{\text{ref}} - E_{\text{aux,ref}}) / (E_{\text{prop,ref}} + E_{\text{stop,ref}} + E_{\text{climb,ref}})$$

The fitted values are E-Transit 1.69, VNR Electric 1.00, eActros 600 1.08. A value near 1 means the physics alone already matches the published range. The van's higher value reflects that its published range is an EPA figure and the 0.45 kWh/km used here is deliberately conservative.

Required energy at departure

$C_{\text{plan}} = C_{\text{nom}} \cdot SOH_{\text{rep}}$	usable capacity as the controller sees it
$E_{\text{req}} = E_{\text{trip}} \cdot (1 + \mu) + \rho_{\text{res}} \cdot C_{\text{plan}}$	μ = 15 % margin, ρ_{res} = 10 % reserve
target SOC = $E_{\text{req}} / C_{\text{plan}}$	capped at 95 % by shortening the route

The margin covers model error, detours and driving style. The reserve means the truck never gets home empty. Both are policy settings. An older battery has a smaller C_{plan}, so the same trip needs a higher target SOC.



Trip alone: 222 kWh (1.46 kWh/km). With margin 33 and reserve 41.
Target 72 % of reported usable 413 kWh (true usable 410 kWh).

Figure. Where the energy goes for T38, a Volvo VNR Electric on a 152 km round trip with 19,518 kg of cargo, 5.5 m of climb per km, at 15 °C. Rolling resistance and climbing dominate because the cargo is heavy.

Sensitivity

Consumption in kWh per km for each model as one input changes, others held at reference. Rows come straight from the model.

CARGO (15 °C, 3 M/KM ASCENT)	EMPTY	50 % CARGO	FULL CARGO
E-Transit	0.38	0.45	0.52
VNR Electric	0.76	1.02	1.28
eActros 600	0.86	1.18	1.50

TEMPERATURE (50 % CARGO, 3 M/KM ASCENT)	−10 °C	0 °C	15 °C	30 °C	40 °C
E-Transit	0.61	0.54	0.45	0.45	0.48
VNR Electric	1.32	1.20	1.02	1.02	1.07
eActros 600	1.47	1.35	1.18	1.18	1.23

TERRAIN (50 % CARGO, 15 °C)	0 M/KM	3 M/KM	6 M/KM	10 M/KM
E-Transit	0.41	0.45	0.49	0.56
VNR Electric	0.85	1.02	1.19	1.41
eActros 600	0.99	1.18	1.37	1.63

5 · WHAT THE TRUCKS TELL THE YARD

SOC and SOH telemetry

The controller never reads a truck's true battery. Each truck reports its state of charge every slot and its state of health once. Both reports are noisy. The physics runs on the true values; the scheduler plans on the reported ones.

$$SOC_{rep} = \text{clip}(\, SOC + \mathcal{N}(0, \sigma_s^2),\, 0,\, 1\,) \qquad \sigma_s = 1\ \%\ \text{absolute}$$

$$SOH_{rep} = SOH + \mathcal{N}(0, \sigma_h^2) \qquad \sigma_h = 1\ \%,\ \text{true SOH spread } 82\text{–}100\ \%$$

$$P_{max}(SOH) = P_{rated} \cdot (1 - 0.5 \cdot (1 - SOH)) \qquad \text{older packs accept less power}$$

$$C_{true} = C_{nom} \cdot SOH \qquad \text{physical capacity}$$

A 1 % SOC error is 0.9 kWh on a van and 5.9 kWh on an eActros. This is why the planner aims slightly above the target (a 2 % cushion) and why the edge agent checks the true energy every minute and adds power if reality falls behind (§10).

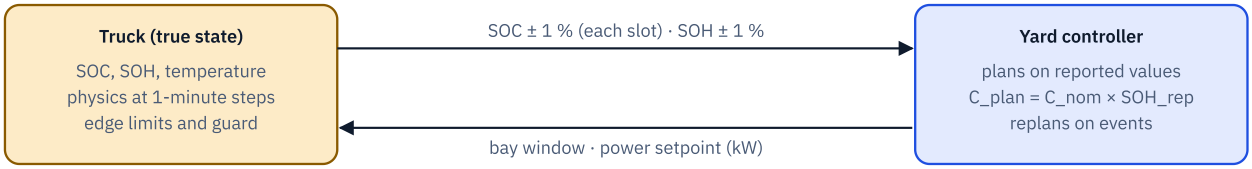


Figure 2. Reports flow up, setpoints flow down. A real deployment would replace the simulated reports with the vehicle's BMS data over the charging protocol.

6 · CHARGING

Charging physics

Charging adds energy at the charger's efficiency, limited by the charger rating, the truck's SOH-derated acceptance, and the way batteries slow down near full.

$$e_{t+1} = e_t + \eta_c \cdot p_t \cdot \Delta t \qquad \eta_c = 0.93,\ \Delta t = 15\ \text{min in the planner, 1 min in the simulator}$$

$$p_t \le P_{max} \cdot (\, \varphi + c_t \cdot (1 - SOC_t)\,),\ c_t = (1 - \varphi) \,/\, (1 - s_0) \qquad \text{taper above } s_0 = 80\ \%,\ \text{floor } \phi = 30\ \%\ \text{of } P_{max}$$

$$p_t + (P_{max}\, c_t \,/\, C) \cdot e_t \le P_{max}\, (\varphi + c_t) \qquad \text{same bound, linear in } e \text{ and } p, \text{ so it fits the LP}$$

In the simulator a thermal state also matters: battery temperature follows the charge rate with a 20-minute time constant, and above 45 °C the edge agent halves the power.

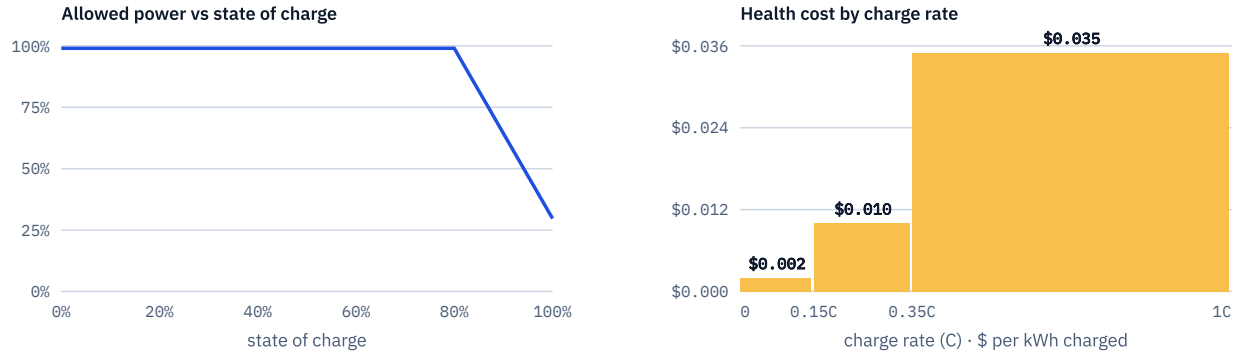


Figure 3. Left: allowed power as a share of the maximum, flat to 80 % SOC then falling to 30 %. Right: the battery-health cost per kWh by charge rate. Costs grow with rate, so the planner prefers many slow kilowatts over few fast ones.

7 · BATTERY CARE

Battery health

A validated ageing model is out of scope for a simulated project. The planner uses a transparent proxy that keeps the problem solvable and captures the two things operators do to protect packs: **charge gently** and **do not sit full**.

$J_{rate} = \Sigma_t \Sigma_k \rho_k \cdot p_{k,t} \cdot \Delta t$, $p_t = \Sigma_k p_{k,t}$ — convex tiers

tier bounds (C-rate): 0.15, 0.35, 1.0 · cost ρ_k ($/kWh): 0.002, 0.010, 0.035

$J_{dwell} = \lambda_h \cdot \Sigma_t \max(0, e_t - 0.8\,C) \cdot \Delta t$ — λ_h = 0.004 $ per kWh above 80 % per hour

Because the tiers are convex, an LP fills the cheap tier first without any integer variables. The dwell term pushes charging towards the end of the window so trucks do not wait fully charged for hours. Both terms are small next to energy cost; their job is to break ties in favour of the battery.

8 · STAGE 2 OF THE SCHEDULER

Power planner (LP)

Given a charging window for each truck, the planner chooses the charge power in every slot. Variables: $p_{v,t}$ (grid-side kW, split across the three rate tiers), $e_{v,t}$ (battery kWh), $h_{v,t}$ *(kWh above 80 %),* s_v *(missed energy),* g_t *(grid import) and* P_{pk} *(peak import).*

$\min \Sigma_t \pi_t g_t \Delta t + \lambda_d P_{pk} + \Sigma_v (J_{rate} + J_{dwell})_v + \Sigma_v M \omega_v s_v$ — energy + demand + health + missed energy

$e_{v,t+1} = e_{v,t} + \eta_c p_{v,t} \Delta t$ — energy balance

$e_{v,0}$ = reported energy now; $e_{v,n} + s_v \geq E_{req,v} + 0.02\, C_{plan,v}$ — reach the target, with a 2 % cushion

$0 \leq p_{v,t} \leq P_{max,v}$(SOH) and taper bound (§6); $p_{v,t}$ = 0 outside the truck's window

$\Sigma_v p_{v,t} - g_t \leq \text{solar}_t$, $g_t \leq P_{pk} \leq$ grid limit — site balance and peak

π_t is the price, λ_d the demand-charge rate, ω_v the truck's priority weight and M = 50 $ per missed kWh. The slack s_v keeps the LP feasible when a deadline cannot be met; the penalty makes missing the deadline the last thing the optimiser accepts, and the shortfall is reported to the operator.

WHY THE DEMAND CHARGE MATTERS

With the demand charge, the planner spreads load and holds site peak to a few hundred kW. Without it, an experiment on the unconstrained model charged more kWh in the cheapest hours (47 % against 24 %) but hit the 1,500 kW grid limit, and the demand charge on that peak cost more than the energy it saved.

9 · THE SCHEDULER

Bay scheduler

With 50 trucks and 11 bays, charging power is not the only decision. Someone has to decide who gets a bay and for how long. The pools are separate: the 6 van bays only take Ford E-Transits, the 3 regional bays only take Volvos, the 2 long-haul bays only take Mercedes trucks.

Anatomy of a bay visit

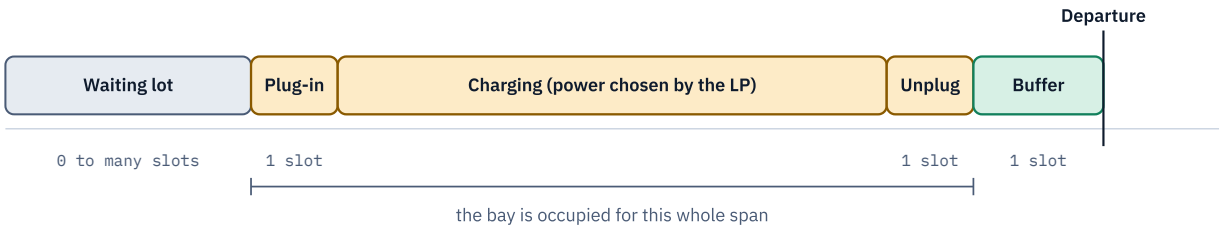


Figure 4. A truck holds a bay from the plug-in slot to the end of the unplug slot. No charging happens in the first or last slot. The last slot before departure is a buffer, so a late connector or a small SOC error does not cause a miss. In the simulator, actual plug-in takes 4–8 min (van) or 8–14 min (tractor), unplug 3–6 or 6–10 min.

Stage 1: sequencing with a time-indexed MILP

Each pool is solved separately. A truck that still needs charge is a job with a release slot r_j (arrival, or now), a deadline D_j (departure minus one buffer slot) and a duration that depends on its own battery:

$$k_j = \min\{\, k : e_k \geq E_{\text{target}} \,\},\quad e_{i+1} = e_i + \eta_c \cdot \min(P_{\max}(\text{SOH}), \text{taper}(e_i)) \cdot \Delta t \qquad \texttt{flat-out slots, with taper}$$

$$p_j = 1 + k_j + 1 \qquad \texttt{plug-in + charge + unplug}$$

Binary $x_{j,t} = 1$ when job j starts at slot t.

$$\min \sum_{j,t} c_{j,t}\, x_{j,t}$$

$$\sum_{t} x_{j,t} = 1 \quad \text{for every job } j \qquad \text{every truck starts exactly once}$$

$$\sum_{j} \sum_{\tau=t-p_j+1 \ldots t} x_{j,\tau} + f_t \le n_b \quad \text{for every slot } t \qquad \text{at most } n_b \text{ bays busy; } f_t \text{ = trucks already docked}$$

$$c_{j,t} = w\, \omega_j\, (t - r_j) + \bar{\pi}_{j,t}\, E_j + M\, \omega_j\, E_{\text{late},j,t} \qquad \text{waiting + energy price + lateness}$$

Here $w = 0.5$ \$ per slot of queueing, $\bar{\pi}_{j,t}$ is the average price over the slots the truck would charge in, E_j its energy, and E_{late} the energy it could not receive if $t + p_j$ overruns the deadline. Capacity constraints have the consecutive-ones structure of interval scheduling, so the model is small and solves fast (about 0.4 s for all three pools). Trucks already in a bay are fixed jobs. If the solver returns nothing, an earliest-deadline-first rule is used.

Stage 1b: widening

Stage 1 books the shortest possible visit. That leaves no room for slow, cheap charging. Stage 1b grows each window, earliest deadline first, into any bay-slots that no one else has booked, in both directions. A truck in a quiet pool might get hours of slack and charge gently on solar power; a truck in a crowded pool gets only what it needs.

Stage 2: power inside the windows

The LP of §8 runs with each truck's window as its allowed charging slots: from the slot after the plug-in slot to the slot before the unplug slot. Sequencing and power are therefore chosen in two steps that each see the other's output: stage 1 sees prices and deadlines, stage 2 sees solar, the site cap, the demand charge and battery health.

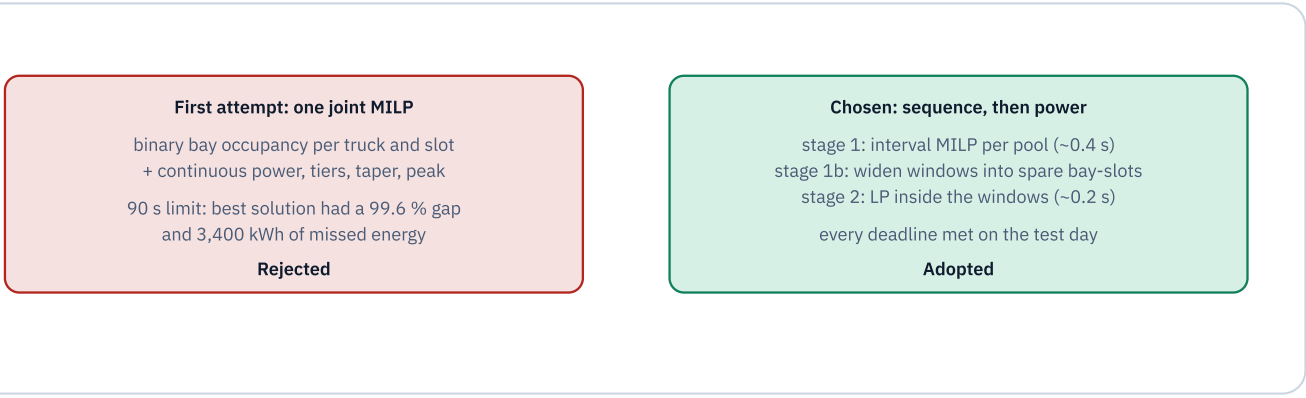


Figure 5. The joint formulation is correct but too hard for a branch-and-bound solver at this size. Splitting it keeps both parts exact and fast, at the price of some optimality.

When it re-plans

Plans are rebuilt on arrivals, on departures of trucks that were not ready, when a truck finishes and frees a bay, when a charger fails or comes back, and at least every four slots. Each re-plan starts from the trucks' reported energy, so the plan always reflects the latest telemetry.

10 · DIGITAL TWIN

Yard simulator

The simulator is the stand-in for the real depot. It owns the true battery state of every truck, runs the physics each minute, and enforces bays and handling delays.

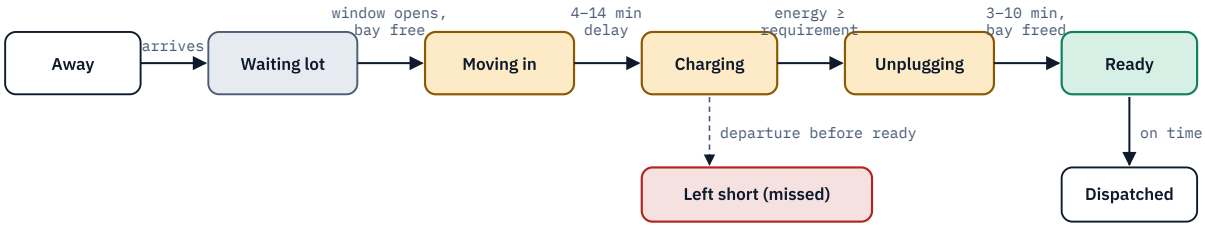


Figure 6. States of a truck in the yard. Delays are drawn per visit from model-specific ranges. A truck counts as on time only if it is in the Ready state, at the gate, when its departure slot starts.

The edge agent

Every minute each charging truck's agent computes the power it will actually draw:

$$p = \min(\, p_{\text{plan}} + p_{\text{guard}},\ \text{taper}(SOC),\ P_{\max}(\text{SOH})\,) \cdot s_{\text{site}} \cdot (0.5 \text{ if } T_{\text{batt}} > 45\ ^\circ\text{C})$$

$$p_{\text{guard}} = \max(0,\ E_{\text{req}} + 0.004\,C - e_{\text{true}} - \eta_c \textstyle\sum_{\text{future}} p_{\text{plan}}\,\Delta t) \,/\, (\eta_c \cdot \text{time to ready-by}) \qquad \text{only when behind plan}$$

$$s_{\text{site}} = \min(1,\ (\text{grid limit} + \text{solar}_{\text{actual}}) \,/\, \Sigma\ \text{requested}) \qquad \text{proportional site cap}$$

The guard is zero while the truck is on track, so it never overrides the plan's decision to wait for cheaper power. It only acts when true energy plus the energy still planned would fall short of the requirement.

Costs counted

Energy cost is price times grid import (load minus realised solar, never below zero). Demand cost is the demand rate times the day's highest 15-minute average import. Battery cost applies the tier and dwell terms of §7 to the true charge rates and SOC. Peak, solar use, waiting time, bay utilisation, overcharge (energy beyond the requirement) and on-time rate are logged per run.

The policies compared

POLICY	WHO GETS A BAY	POWER	ROLE
First come, first served	Earliest arrival in its pool	Flat out until the requirement is met	What most depots do
Earliest departure first	Earliest deadline in its pool	Flat out	Standard scheduling rule
Smart scheduler	MILP windows (§9)	LP inside the window, edge guard on top	This project
Ideal	No bays: a charger per truck	LP, no handling delays	A bound, not a real option

11 · LANGUAGE AND SEARCH

Nemotron and Tavily

NVIDIA Nemotron models run on Nebius Token Factory through its OpenAI-compatible endpoint. Three jobs use them. None of them changes a charge setpoint or a bay.

JOB	MODEL	WHAT IT DOES	SAFEGUARD
Dispatcher notes to actions	`nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B`	"Truck 7 is running two hours late. T12's route is now 200 km." becomes structured actions (delay, route, cargo kg, departure time, charger fault)	Actions are validated against known truck IDs and action types. A regex parser takes over if the call fails or returns nothing valid.
Live temperature	Nano, with Tavily search	Tavily returns forecast text for the depot city; Nemotron extracts the Celsius value; the energy model recomputes every truck's requirement	Returns null instead of guessing; values outside −45…55 °C are rejected; default 15 °C if the lookup fails
Explanations	`nvidia/nemotron-3-super-120b-a12b`	Answers "why did T20 wait and charge when it did?" using only a JSON fact sheet: the energy breakdown, wait time, charging windows, prices	Fact-only prompt; a templated explanation is the fallback

Model IDs were taken from the account's own model list. It also serves `nvidia/Nemotron-3-Ultra-550b-a55b`, which is the candidate for harder explanation requests. Tavily is a runtime dependency of the weather feature, which qualifies for the Tavily bonus prize.

What the live calls taught us

- The Nano model parsed a three-sentence dispatcher note into the right three actions in 7.5 s. Super answered an explanation in 3.7 s.
- These models spend tokens on reasoning before answering. With a 500-token limit an explanation was cut mid-sentence. The client now allows 1,500 to 2,500 tokens and treats `finish_reason: length` as a failure to be handled instead of returning half an answer.
- When Tavily returned no forecast, an early prompt made the model answer 0 °C, which silently inflated every truck's energy by 16 %. The prompt now allows null, and the code validates the range.
- The model's list endpoint spells the Nano ID with different capitalisation from the one found in search results. Always read the list.

12 · MEASURED

Results

Eight simulated depot-days (seeds 0–7), four policies each, with realistic delays and noisy telemetry. All trucks in a run share the same random day across policies, so the comparison is paired.

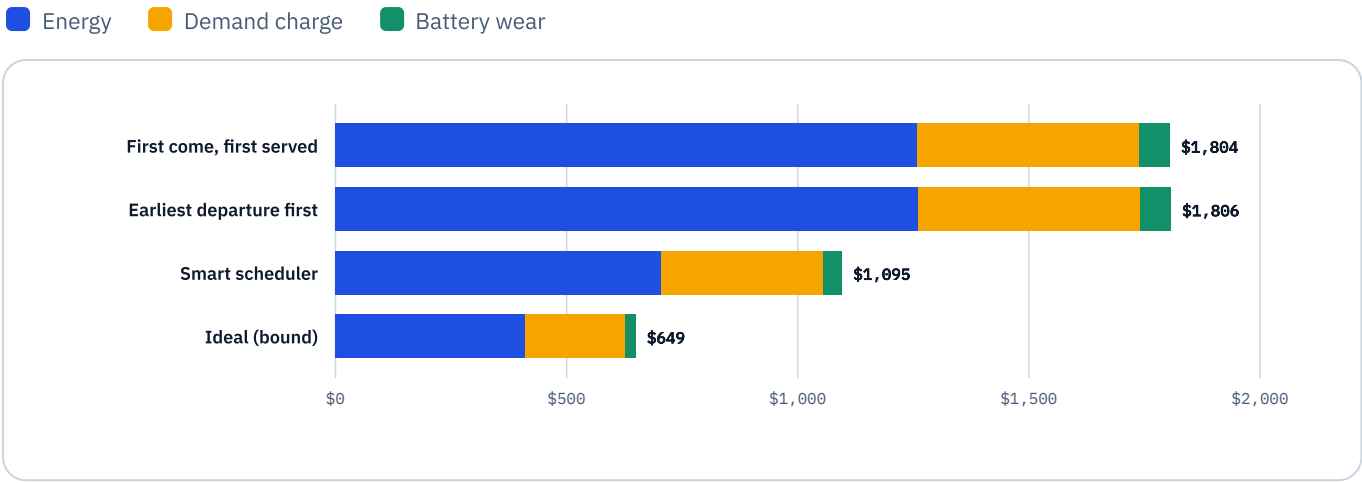


Figure 7. Average daily cost over 8 days. The smart scheduler is about 39 % below first-come-first-served. The ideal bound shows what bays and handling delays cost: that gap is the room left for a better scheduler.

METRIC	FIRST COME, FIRST SERVED	EARLIEST DEPARTURE FIRST	SMART SCHEDULER	IDEAL (BOUND)
Total cost per day	$1,804 ±91	$1,806 ±91	$1,095 ±40	$649 ±135
Energy	$1,259 ±88	$1,261 ±89	$705 ±30	$411 ±70
Demand charge	$480 ±8	$480 ±10	$349 ±28	$216 ±79
Battery wear	$66 ±4	$65 ±4	$41 ±2	$22 ±2
Peak grid import (kW)	872 ±14	872 ±17	635 ±51	392 ±143
Missed departures per day	0.00	0.00	0.00	0.00
Mean wait for a bay (h)	1.2 ±0.3	1.2 ±0.2	2.7 ±0.4	0.0
Longest wait (h)	3.9 ±0.7	6.4 ±0.7	11.0 ±0.7	0.0
Bay utilisation	40% ±2%	40% ±2%	67% ±4%	0%
Solar used on site	29% ±2%	29% ±2%	31% ±4%	50% ±5%
Energy beyond requirement (kWh)	24.5 ±3.7	24.5 ±3.7	16.2 ±2.6	14.2 ±2.7
Average charge rate (C)	0.29 ±0.00	0.29 ±0.00	0.23 ±0.01	0.17 ±0.01
Truck-hours above 80 % charge	241 ±26	238 ±24	127 ±24	50 ±29
Energy charged above 0.15 C	100% ±0%	100% ±0%	74% ±4%	28% ±4%
Planner runs per day	0	0	77	69
Total solver time per day (s)	0.0	0.0	15.0 ±2.2	9.0 ±1.3

BATTERY STRESS

The last three rows are why the smart scheduler is gentler on the packs, not only cheaper. It charges at a lower average rate and leaves trucks sitting full for far fewer hours, because it finishes charging close to departure instead of on arrival.

READING THE TABLE

The smart scheduler makes trucks wait longer (about 2.7 h on average against 1.2 h) because it holds them until the charging window is cheap. Waiting in the lot costs nothing as long as the truck is ready by dispatch. Bay utilisation rises from 40 % to 67 % because trucks charge slowly while they hold a bay.

One day in detail (seed 0)

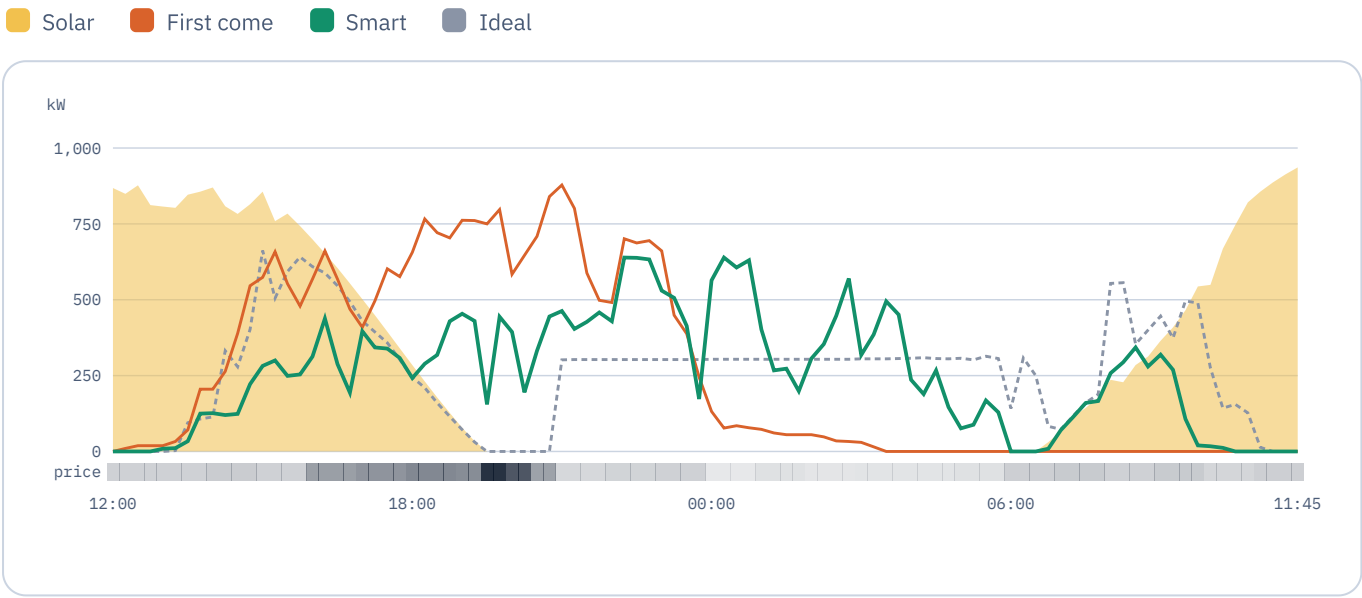


Figure 8. Charging load through the day. First come, first served charges flat out on arrival and peaks above 850 kW. The smart scheduler flattens the peak and moves energy into cheap and solar hours. The strip under the axis shows the price (darker is dearer).

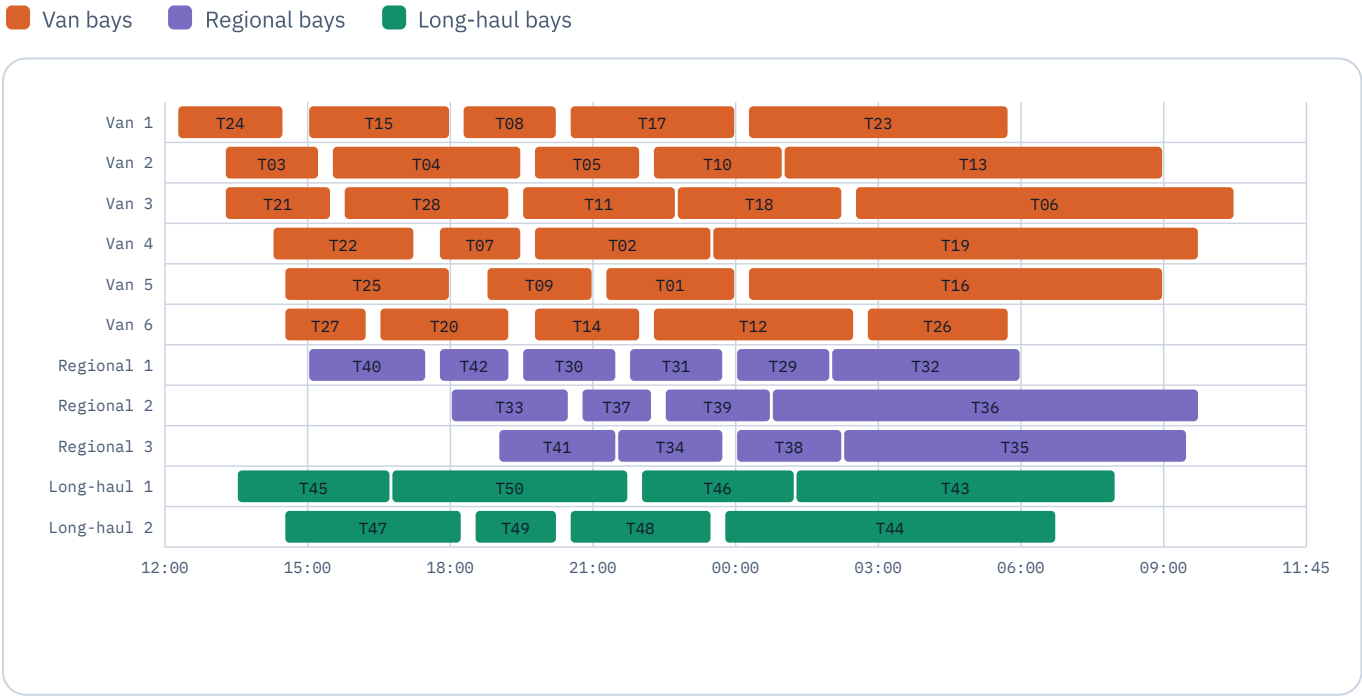


Figure 9. Which truck holds which bay, when, under the smart scheduler. Each bar is one visit, labelled with the truck. Bars are longer than a flat-out charge would need because the scheduler uses spare bay time to charge gently.

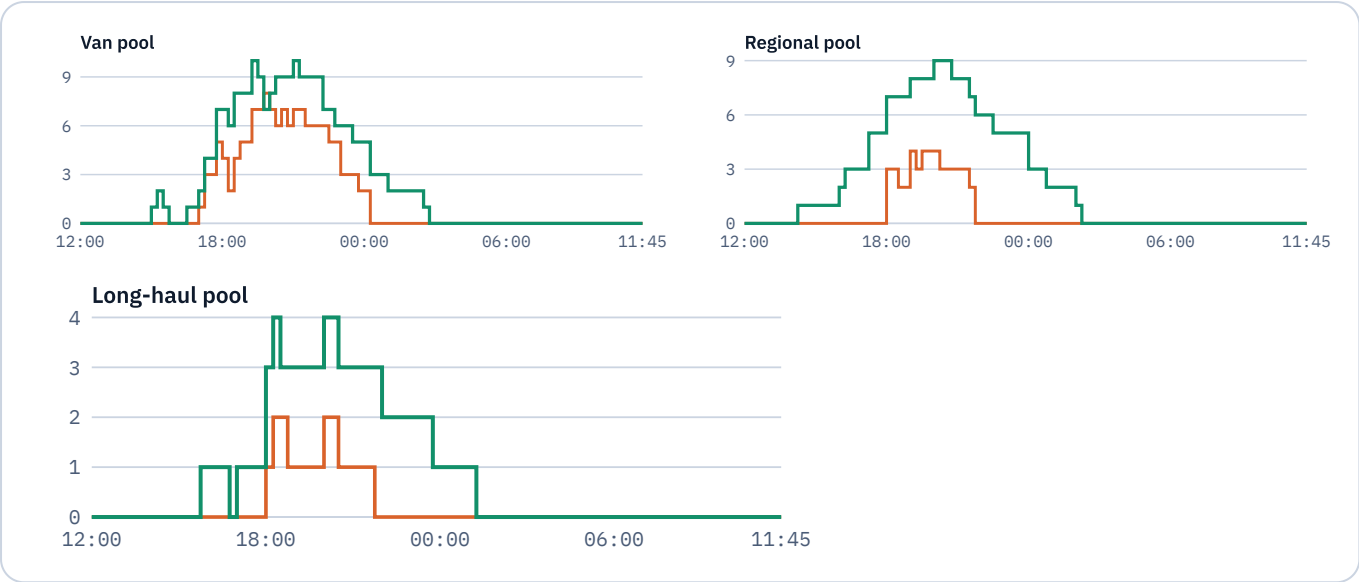


Figure 10. Trucks waiting for a bay, per pool, first come (orange) against smart (green). The smart scheduler keeps more trucks waiting for longer; they are not late, they are waiting for a cheaper window.

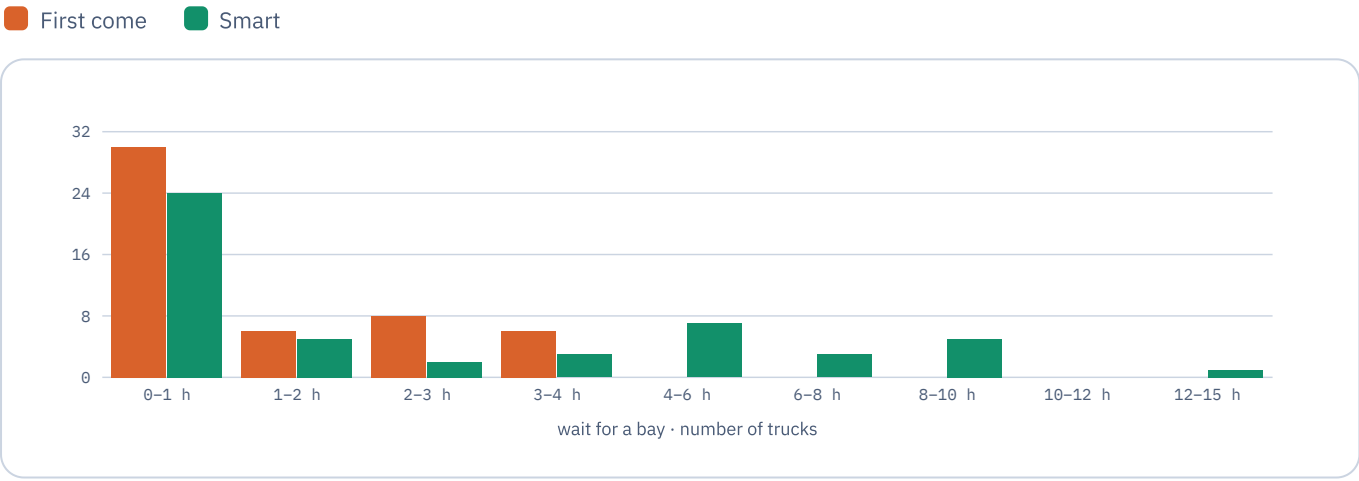


Figure 11. How long each of the 50 trucks waited for a bay on seed 0. The long tail under the smart scheduler is deliberate and bounded by each truck's deadline.

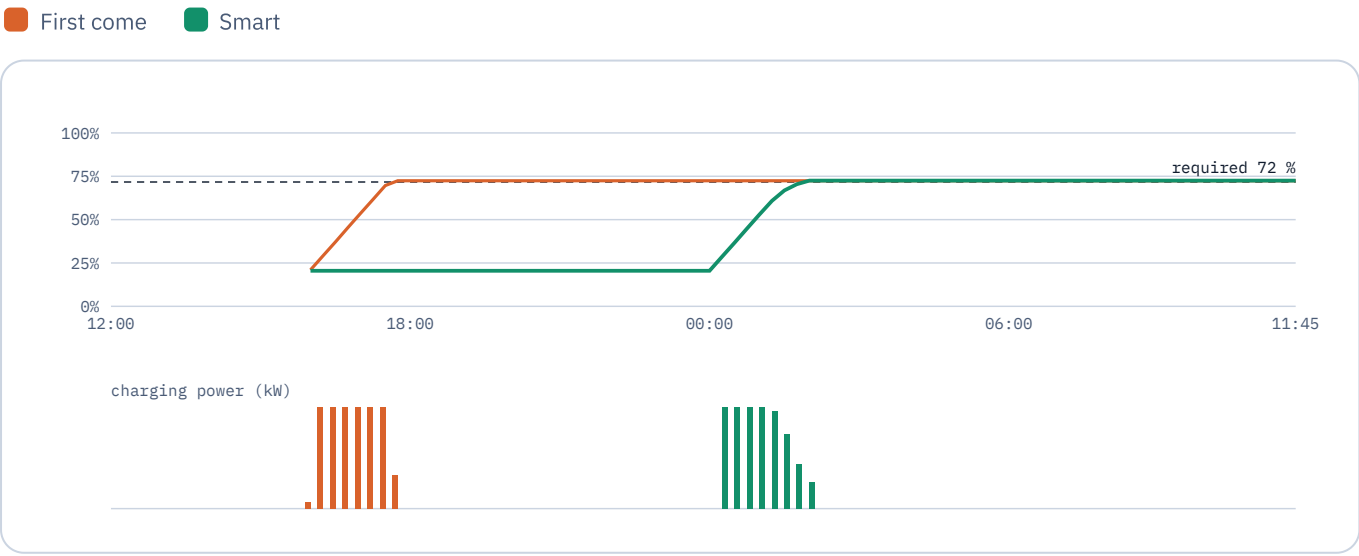


Figure 12. T38 (Volvo VNR Electric, 19,518 kg cargo, true SOH 91 %, reported 91 %). It arrives at 16:00 with 21 % and leaves at 09:00. First come, first served charges at full power from 16:00; the smart scheduler charges between 00:15 and 02:15 at a lower rate. Both reach the required level.

Stress: cold weather with charger faults

Six days at −10 °C, with each charging truck facing a 0.4 % chance per slot of its connector failing for two hours.

METRIC	FIRST COME, FIRST SERVED	EARLIEST DEPARTURE FIRST	SMART SCHEDULER	IDEAL (BOUND)
Total cost per day	$1,937 ±107	$1,935 ±108	$1,305 ±102	$802 ±109
Energy	$1,372 ±108	$1,373 ±111	$846 ±88	$525 ±76
Demand charge	$471 ±13	$468 ±23	$393 ±32	$247 ±37
Battery wear	$94 ±5	$94 ±4	$66 ±5	$30 ±4
Peak grid import (kW)	857 ±24	851 ±42	714 ±58	448 ±67
Missed departures per day	0.50 ±0.55	0.00	0.50 ±0.84	0.17 ±0.41
Mean wait for a bay (h)	1.9 ±0.1	2.0 ±0.1	3.6 ±0.3	0.0
Longest wait (h)	5.6 ±0.6	9.6 ±1.5	12.6 ±1.4	0.0
Bay utilisation	51% ±2%	51% ±2%	71% ±3%	0%
Solar used on site	32% ±2%	32% ±2%	36% ±7%	53% ±3%
Energy beyond requirement (kWh)	20.5 ±1.8	20.5 ±1.8	15.2 ±1.6	13.6 ±1.2
Average charge rate (C)	0.28 ±0.00	0.28 ±0.00	0.24 ±0.00	0.17 ±0.01
Truck-hours above 80 % charge	390 ±35	388 ±35	266 ±37	99 ±30
Energy charged above 0.15 C	99% ±0%	99% ±0%	81% ±2%	30% ±4%
Planner runs per day	0	0	77	74
Total solver time per day (s)	0.0	0.0	18.3 ±1.9	10.2 ±1.0

What does not work yet

- **Occasional misses.** The smart scheduler made every departure on all 8 normal days and missed 0.5 trucks per day in the stress runs. The stress misses follow connector faults that remove hours of charging from a truck with a tight deadline. A fault-aware priority bump and earlier re-planning are the next fixes.
- **The ideal bound is not immune.** It had no misses on the 8 normal days but 0.17 per day in the stress runs. It plans to the wire, and noise in the SOC reports can leave a truck a few kWh short. That is the effect the planning cushion exists for.
- **Solar use is modest for every policy** (31 % for the smart scheduler, 50 % even for the ideal bound) because most trucks arrive after the sun is going down. Using more solar needs daytime dwell time, which this fleet does not have.

13 · INTERFACES

Interfaces

Web application

`python server.py` serves the interface at `localhost:8000` and runs the real scheduler and simulator behind it. It simulates all four policies for a chosen seed, temperature and fault rate, then shows:

- **The yard.** Bays with trucks whose trailers fill as the battery charges, a time-to-target line on each card (for example "1h 15m to 85%"), per-pool queues with slack chips, and a ready lane. Filters highlight the trucks at risk of missing departure or above 80 % charge and dim the rest.
- **Time navigation.** A scrubber marked with the peak-tariff window, solar peak, arrival wave and departure wave, with one-click jumps to each, to the current time of day, and play at 1x, 2x or 4x.
- **Site power.** Solar, charging load and grid import against the grid limit on the left axis and price on a right-hand axis. The headroom between grid import and the limit is shaded wherever import comes within 30 % of the limit, and the day's peak is labelled as the value that sets the demand charge.
- **Bay schedule.** Who holds which bay and when. Hovering a bar shows arrival charge, target, power and deadline. A cursor follows the scrubber across the schedule, the queue charts and the site power chart.
- **Policy comparison**, including a battery-stress panel: average charge rate, truck-hours above 80 % charge, share of energy delivered above 0.15 C, and the wear cost.
- **Truck drawer** with the energy breakdown, charge curve, timeline and a Nemotron explanation.
- **Dispatcher drawer** with recommended actions, quick scenarios and free-text notes (next section).

The layout adapts from wide desktop monitors down to 1,024 px tablets: grids reflow, the queue and ready lane move under the bays, and nothing scrolls sideways.

Yard design

The yard view follows a design prototype: each charging bay holds a truck drawn side-on whose trailer fills as the battery charges; a dashed line marks the charge the trip needs; amber means charging, green means ready, slate means waiting. Each pool has a queue with a position, wait time and a slack chip (plenty, tight, at risk), and a ready lane. The application renders it from the scheduler's real output, including the plug-in and unplug states.

Dispatcher console

The dispatcher drawer has three parts.

- **Recommended actions.** When a truck misses its departure, cannot be charged for its trip, or a charger fails, the drawer proposes concrete fixes with numbers computed from the same models: a delay long enough to finish the charge, the longest route its delivered energy supports, a cargo weight that fits the battery, or marking a faulted charger repaired. One click applies the change and re-plans. For a charger fault it also names which waiting truck to dock in the held bay.
- **Quick scenarios.** Grid curtailment (import limit cut to 900 kW), a cold snap (−10 °C), a 3-hour route extension for the regional tractors (held to what each battery can carry), and a reset.
- **Free-text notes.** "Truck 5 is carrying 1.6 tonnes of cargo. Truck 30 route is now 280 km. Truck 7 delayed 2 hours." become actions; the energy model recomputes each truck's requirement; the scheduler re-plans. Trips that exceed the battery are flagged immediately.

14 · QUALITY

Testing

Twelve automated tests run in about 30 seconds (`python -m pytest -q`):

- The fleet has 50 trucks of all three models.
- The energy model reproduces each model's rated consumption at reference conditions.
- Energy rises with distance, cargo and terrain and cold weather, and falls with lighter cargo.
- The requirement is trip energy plus the margin plus the reserve, exactly.

- The unconstrained LP meets every deadline within the site limit.
- Stage 1 windows never exceed a pool's bay count and always sit between arrival and the dispatch buffer.
- The full scheduler delivers the energy and never charges in a plug-in slot or after a window.
- An impossible deadline is reported as a shortfall, not hidden.
- The dispatcher parser handles delays, route changes, cargo weights and faults.
- An end-to-end day: everyone dispatched, handling delays present, costs ranked ideal < smart < first-come, and no pool above its bay count at any minute.

15 · HONEST LIMITS

Limits and assumptions

- **Everything is simulated.** Prices, solar, arrivals, trips, cargo and SOH are generated. Truck battery and charging figures are public; mass, drag and drivetrain parameters are estimates, tied to the real range by calibration.
- **The energy model is physics-flavoured, not a vehicle model.** It has no route-level speed profile, wind, traffic or tyre data. The 15 % margin exists because of this.
- **Battery health is a proxy.** The tiers and dwell cost are plausible and convex, not fitted to cell ageing data. SOH only changes capacity and power limit; it does not evolve during the day.
- **Telemetry noise is Gaussian and unbiased.** Real BMS estimates can be biased, delayed or missing.
- **No uncertainty in the plan.** The planner is deterministic with a fixed cushion. A scenario-based or robust planner could use less margin.
- **Bay assignment is per-pool and not preemptive.** A truck in a bay stays until it is done. Moving a half-charged truck out for an urgent one is not modelled.
- **No vehicle-to-grid, no site battery, no export.** Surplus solar is curtailed.
- **No real charger protocol.** The edge agents are simulated; a real deployment would use OCPP or the vendor's interface.
- **Handling delays are drawn from uniform ranges** chosen to be plausible, not measured.

16 · SUBMISSION

Hackathon fit

JUDGING CRITERION	HOW THE PROJECT ANSWERS IT
Technological implementation	Two-stage exact scheduler (MILP then LP) on HiGHS, a physics energy model, noisy-telemetry planning, 1-minute edge agents, 12 tests; Nemotron on Token Factory in the loop with validated outputs and fallbacks
Design	Web application built from the design: trailers that fill as batteries charge, per-pool queues with risk chips, a ready lane, a bay schedule and a truck drawer with the energy waterfall
Potential impact	Depots with fewer chargers than vehicles pay for peak power and idle bays; the benchmark shows about 39 % lower daily cost and 27 % lower peak than first come, first served, in simulation
Quality of the idea	Treats the charger bay as the scarce resource and the battery's real requirement (cargo, terrain, temperature, age) as the target; uses the language model for intake and explanation only, not for decisions

Services used

- **Nebius Token Factory**: Nemotron Nano (parsing, weather extraction) and Super (explanations), called at runtime.
- **NVIDIA Nemotron** open models: `NVIDIA-Nemotron-3-Nano-30B-A3B` and `nemotron-3-super-120b-a12b`; Ultra is available for later.
- **Tavily**: live weather search feeding the energy model.
- **Nebius Serverless Jobs**: the benchmark script and Dockerfile are ready (`scripts/benchmark.py`). A large cloud run has not been done yet and is optional.

Feedback notes for the submission form

- Token Factory's OpenAI-compatible API worked with a plain HTTP client on the first try, and the model-list endpoint made ID discovery simple.
- Reasoning tokens count against `max_tokens` . A documented recommendation for output budgets on the Nemotron 3 models would save time.
- Model naming differs in capitalisation between public pages and the API list.
- For Tavily, the answer field is concise and good, but forecasts for a specific day are often missing; extraction must handle "no data".

Still to do before the deadline (30 Oct 2026)

1. Create the public GitHub repository (MIT license is in place) and add the license and README to the top page.
2. Record the 3-minute demo video: yard replay, dispatcher note, energy waterfall, Nemotron explanation, benchmark.
3. Optionally run the benchmark as a Nebius Serverless Job for a 100-seed result.
4. Fix the fault-related misses (§12) and rotate every API key that was used during development.
5. Fill the Devpost form: track, description, demo, video, repository, feedback.

17 · CODE

Repository

FILE	CONTENTS
`fleetcharge/specs.py`	Truck models (battery, charging, mass, drag, delays), bays, site, health, dispatch and telemetry settings
`fleetcharge/energy.py`	Physics trip-energy model with calibration and breakdown
`fleetcharge/scenario.py`	Price, solar and fleet generation; requirement and target; reported SOC and SOH
`fleetcharge/planner.py`	Power LP
`fleetcharge/scheduler.py`	Bay sequencing MILP, window widening, full scheduler
`fleetcharge/yard.py`	Yard digital twin, edge agents, four policies
`fleetcharge/agent.py`	Nemotron client, dispatcher parser, Tavily weather, explanations
`server.py` , `web/`	Python server (stdlib) with the JSON API, and the yard web interface
`scripts/benchmark.py` , `Dockerfile`	Multi-seed benchmark, ready for a Serverless Job
`tests/test_core.py`	The 12 tests

Run it

```
pip install -r requirements.txt
export NEBIUS_API_KEY=...      # optional, offline fallbacks otherwise
export TAVILY_API_KEY=...      # optional
python server.py               # http://localhost:8000
python -m pytest -q
python scripts/benchmark.py --seeds 20 --workers 8
python scripts/benchmark.py --seeds 20 --temp -10 --fault-rate 0.004
```

APPENDIX

Parameters

SETTING	VALUE	MEANING
Slot length	15 min	Planner step; the simulator steps every minute
Energy margin μ	15 %	On top of the trip energy
Reserve	10 % of usable	Arrival buffer
Target cap	95 %	Longest trip dispatch will assign
Dispatch buffer	1 slot	Truck must be ready this long before departure
Planning cushion	2 % of usable	Extra energy the LP aims for, against SOC noise
Guard goal	+0.4 % of usable	Edge agent aims slightly above the requirement
Missed-energy penalty	50 $/kWh	Times the truck's priority weight (1 or 2)
Queue weight w	0.5 $/slot	Per unit priority, stage 1
Bay holding cost	0.35 $/slot	Configured; used by the joint model that was rejected
SOC / SOH noise	1 % / 1 %	One sigma, absolute
Replan interval	≥ every 4 slots	Plus on events

Built for the Nebius x NVIDIA Global AI Hackathon. Simulated data throughout.